

第五章 时变电磁场

时变电磁场：随时间变化的电场和磁场

时变电磁场的方程

边界条件

波动方程

位函数

位函数求解

时变场的能量与功率

正弦时变电磁场

正弦电磁场的能量与功率

5.1 麦克斯韦(Maxwell)方程

1) 电磁场方程

$$\oiint_S \vec{D} \cdot d\vec{S} = q$$

$$\nabla \cdot \vec{D} = \rho$$

$$\vec{D} = \epsilon \vec{E}$$

$$\oint_l \vec{E} \cdot d\vec{l} = - \iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\vec{J} = \sigma \vec{E}$$

$$\oiint_S \vec{B} \cdot d\vec{S} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\vec{B} = \mu \vec{H}$$

$$\oint_l \vec{H} \cdot d\vec{l} = I$$

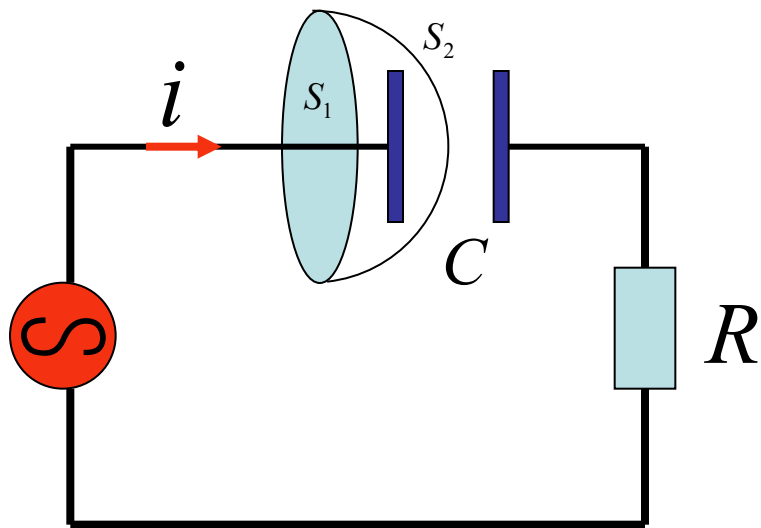
$$\nabla \times \vec{H} = \vec{J}$$

$$\oint_S \vec{J} \cdot d\vec{S} = 0$$

$$\nabla \cdot \vec{J} = 0$$

2) 位移电流

$$\oint_l \vec{H} \cdot d\vec{l} = I = \iint_s \vec{J} \cdot d\vec{S} = \dot{i} = \begin{cases} i, S_1 \\ 0, S_2 \end{cases}$$



为什么会出现这一矛盾的结果呢?

$$\nabla \times \vec{H} = \vec{J}$$

$$\nabla \cdot \nabla \times \vec{H} = 0$$

$$\nabla \cdot \vec{J} = 0$$

$$\nabla \cdot \nabla \times \vec{H} = \nabla \cdot \vec{J}$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t} \neq 0$$

$$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot (\vec{J} + \frac{\partial \vec{D}}{\partial t}) = 0$$

$$\nabla \cdot \vec{D} = \rho$$

位移电流密度

$$\oint_l \vec{H} \cdot d\vec{l} = I + \iint_s \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$$

位移电流

位移电流也是磁场的涡旋源
变化的电场产生磁场

3) Maxwell 方程(组)

$$\oint_l \vec{H} \cdot d\vec{l} = I + \iint_s \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$$

全电流安培环路定律

$$\oint_l \vec{E} \cdot d\vec{l} = -\iint_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

法拉第电磁感应定律

$$\oiint_s \vec{D} \cdot d\vec{S} = q$$

电场高斯定理

$$\oiint_s \vec{B} \cdot d\vec{S} = 0$$

磁通连续性原理

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{D} = \epsilon \vec{E}$$

变化的电场产生磁场,全电流是磁场的旋度源

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{J} = \sigma \vec{E}$$

变化的磁场产生电场,是电场的旋度源

$$\nabla \cdot \vec{D} = \rho$$

电荷是电场的散度源

$$\vec{B} = \mu \vec{H}$$

$$\nabla \cdot \vec{B} = 0$$

物质结构方程

磁场是无散场,磁力线总是闭合的

Maxwell 各方程的名称

Maxwell 方程的物理意义

变化的磁场产生电场,变化的电场产生磁场
时变电场与磁场交替产生,不可分割
在空间形成电磁波

4) Maxwell方程的独立性

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{D} = \epsilon \vec{E}$$

$$\vec{E} \quad \vec{D} \quad \vec{H} \quad \vec{B} \quad \vec{J} \quad \rho$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{J} = \sigma \vec{E}$$

16个标量

$$\nabla \cdot \vec{D} = \rho$$

$$\vec{B} = \mu \vec{H}$$

有2个方程不独立

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

18个标量方程

5.2 时变电磁场边界条件

1) 两种一般媒质的边界条件

$$\oiint_S \vec{B} \cdot d\vec{S} = 0 \quad \Rightarrow \quad B_{1n} = B_{2n}$$

$$\oiint_S \vec{D} \cdot d\vec{S} = q \quad \Rightarrow \quad D_{1n} - D_{2n} = \rho_s$$

$$\oint_l \vec{E} \cdot d\vec{l} = - \iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad \Rightarrow \quad E_{1t} = E_{2t}$$

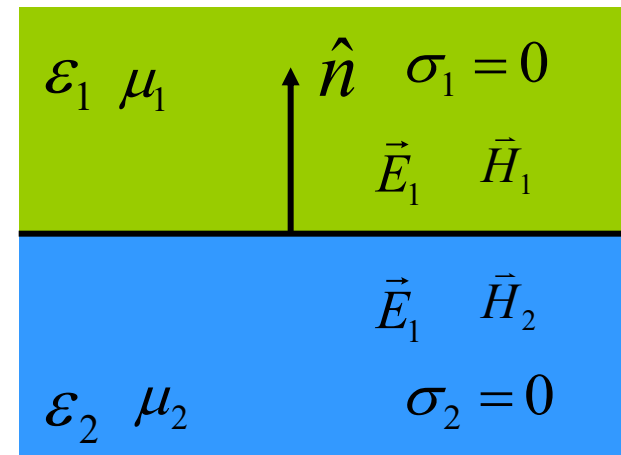
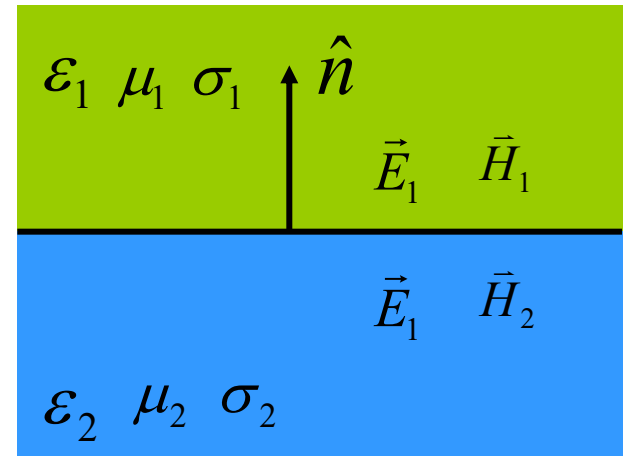
$$\oint_l \vec{H} \cdot d\vec{l} = I + \iint_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S} \quad \Rightarrow \quad H_{1t} - H_{2t} = J_s$$

2) 两种理想介质的边界条件

$$\sigma_1 = 0 \quad \sigma_2 = 0 \quad \rho_s = 0 \quad J_s = 0$$

$$B_{1n} = B_{2n} \quad D_{1n} = D_{2n}$$

$$E_{1t} = E_{2t} \quad H_{1t} = H_{2t}$$



3) 理想介质与理想导体的边界条件

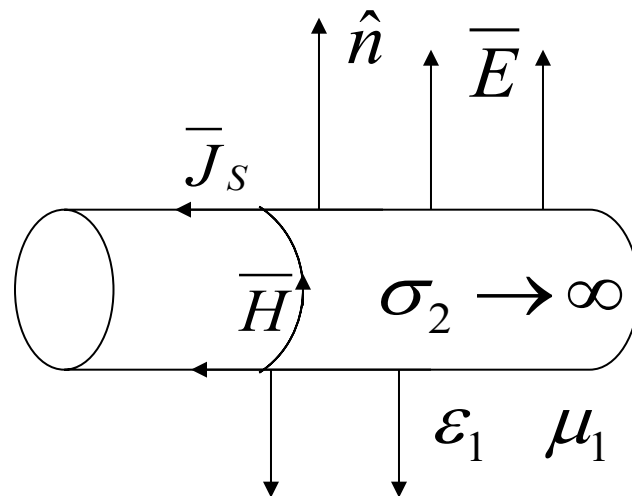
- 时变场在理想导体内部，电磁场和电流均为零。
- 在理想导体表面

$$\hat{n} \cdot (\bar{D}_1 - \bar{D}_2) = \rho_s \quad \hat{n} \cdot \bar{D}_1 = \rho_s$$

$$\hat{n} \times (\bar{E}_1 - \bar{E}_2) = 0 \quad \hat{n} \times \bar{E}_1 = 0$$

$$\hat{n} \cdot (\bar{B}_1 - \bar{B}_2) = 0 \quad \hat{n} \cdot \bar{B}_1 = 0$$

$$\hat{n} \times (\bar{H}_1 - \bar{H}_2) = \bar{J}_s \quad \hat{n} \times \bar{H}_1 = \bar{J}_s$$



5.3 波动方程及位函数

1) 波动方程

$$\nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t} = -\mu \frac{\partial \bar{H}}{\partial t}$$

$$\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t} = \bar{J} + \varepsilon \frac{\partial \bar{E}}{\partial t}$$

$$\nabla \times \nabla \times \bar{E} = -\nabla \times \mu \frac{\partial \bar{H}}{\partial t}$$

$$\nabla \cdot \bar{D} = \rho \quad \nabla \cdot \bar{E} = \frac{\rho}{\varepsilon}$$

$$= -\mu \frac{\partial}{\partial t} (\nabla \times \bar{H}) = -\mu \frac{\partial}{\partial t} \bar{J} - \mu \varepsilon \frac{\partial^2}{\partial t^2} \bar{E}$$

$$\nabla \times \nabla \times \bar{E} = \nabla (\nabla \cdot \bar{E}) - \nabla^2 \bar{E} = \nabla \frac{\rho}{\varepsilon} - \nabla^2 \bar{E} = \frac{1}{\varepsilon} \nabla \rho - \nabla^2 \bar{E}$$

$$-\mu \frac{\partial}{\partial t} \bar{J} - \mu \varepsilon \frac{\partial^2}{\partial t^2} \bar{E} = \frac{1}{\varepsilon} \nabla \rho - \nabla^2 \bar{E}$$

$$\nabla^2 \bar{E} - \mu \varepsilon \frac{\partial^2}{\partial t^2} \bar{E} = \frac{1}{\varepsilon} \nabla \rho + \mu \frac{\partial}{\partial t} \bar{J}$$

非齐次电场矢量波动方程

磁场波动方程

$$\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$$

$$\nabla \times \nabla \times \bar{H} = \nabla \times \left(\bar{J} + \varepsilon \frac{\partial \bar{E}}{\partial t} \right)$$

$$= \nabla \times \bar{J} + \varepsilon \frac{\partial}{\partial t} \nabla \times \bar{E}$$

$$= \nabla \times \bar{J} - \varepsilon \mu \frac{\partial^2 \bar{H}}{\partial t^2}$$

$$\nabla \times \nabla \times \bar{H} = \nabla(\nabla \cdot \bar{H}) - \nabla^2 \bar{H} = -\nabla^2 \bar{H}$$

$$\nabla^2 \bar{H} - \mu \varepsilon \frac{\partial^2}{\partial t^2} \bar{H} = -\nabla \times \bar{J}$$

$$\nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t} = -\mu \frac{\partial \bar{H}}{\partial t}$$

$$\nabla \cdot \bar{B} = 0 \quad \nabla \cdot \bar{H} = 0$$

非齐次磁场矢量波动方程

2) 位函数

$$\nabla \cdot \vec{B} = 0 \quad \Rightarrow \quad \vec{B} = \nabla \times \vec{A} \quad \vec{A} \text{ 矢量磁位}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \times \vec{E} = -\frac{\partial}{\partial t} \nabla \times \vec{A} \quad \nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t} \right) = 0$$

$$\nabla \times \vec{H} = \vec{J} + \varepsilon \frac{\partial \vec{E}}{\partial t} \quad \vec{E} + \frac{\partial \vec{A}}{\partial t} = -\nabla \Phi \quad \Phi \text{ 标量电位}$$

$$\nabla \times \vec{B} = \mu \vec{J} + \varepsilon \mu \frac{\partial \vec{E}}{\partial t} \quad \vec{E} = -\nabla \Phi - \frac{\partial \vec{A}}{\partial t}$$

$$\nabla \times \nabla \times \vec{A} = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu \vec{J} - \varepsilon \mu \frac{\partial}{\partial t} \left(\nabla \Phi + \frac{\partial \vec{A}}{\partial t} \right)$$

$$\nabla^2 \vec{A} - \mu \varepsilon \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J} + \nabla \left(\nabla \cdot \vec{A} + \mu \varepsilon \frac{\partial \Phi}{\partial t} \right) \quad \nabla^2 \vec{A} - \mu \varepsilon \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J}$$

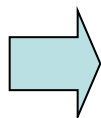
$$\nabla \cdot \vec{A} = -\mu \varepsilon \frac{\partial \Phi}{\partial t} \quad \text{洛伦兹规范条件}$$

标量电位方程

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon}$$

$$\vec{E} = -\nabla\Phi - \frac{\partial \vec{A}}{\partial t}$$



$$\nabla \cdot \left(-\nabla\Phi - \frac{\partial \vec{A}}{\partial t} \right) = \frac{\rho}{\varepsilon}$$

$$\nabla \cdot \vec{A} = -\mu\varepsilon \frac{\partial \Phi}{\partial t}$$

达朗贝尔方程

$$\left\{ \begin{array}{l} \nabla^2 \Phi - \mu\varepsilon \frac{\partial^2 \Phi}{\partial t^2} = -\frac{\rho}{\varepsilon} \\ \nabla^2 \vec{A} - \mu\varepsilon \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J} \end{array} \right.$$

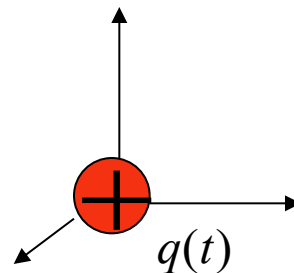
5.4 滞后位--位函数的解

$$\nabla^2 \vec{A} - \mu\epsilon \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J}$$

$$\nabla^2 A_x - \mu\epsilon \frac{\partial^2 A_x}{\partial t^2} = -\mu J_x$$

$$\nabla^2 \Phi - \mu\epsilon \frac{\partial^2 \Phi}{\partial t^2} = -\frac{\rho}{\epsilon}$$

$$\nabla^2 \Phi(r, t) - \mu\epsilon \frac{\partial^2 \Phi(r, t)}{\partial t^2} = -\frac{q(t)\delta(r)}{\epsilon}$$



$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) - \mu\epsilon \frac{\partial^2 \Phi}{\partial t^2} = 0, \quad r \neq 0$$

设: $v = \frac{1}{\sqrt{\mu\epsilon}}$

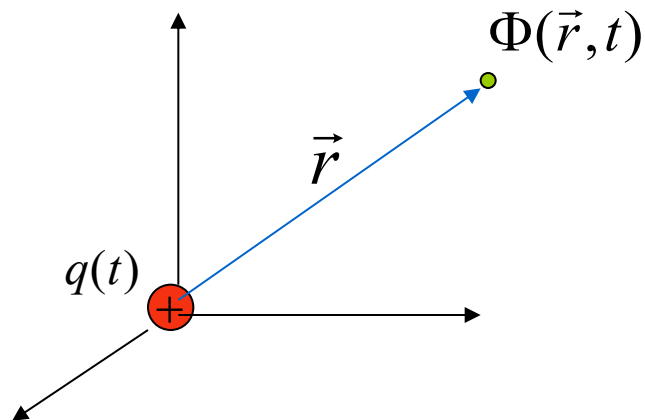
$$\Phi = \frac{u}{r}$$

$$\frac{\partial^2 u}{\partial r^2} - \frac{1}{v^2} \frac{\partial^2 u}{\partial t^2} = 0$$

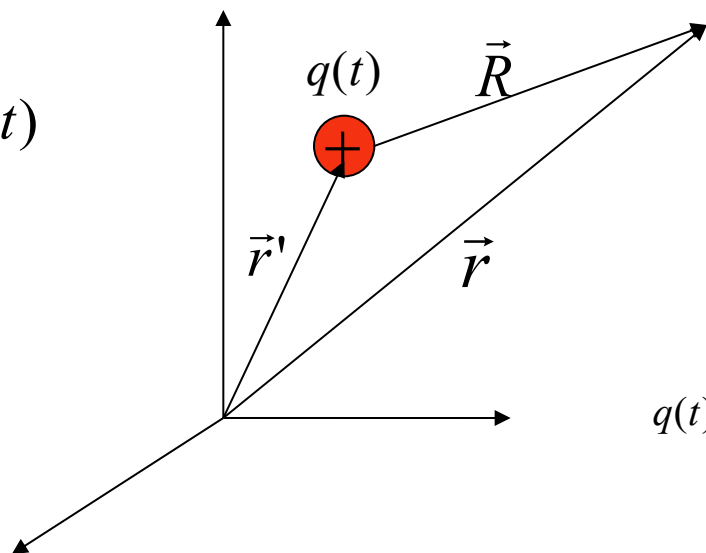
$$u(r, t) = c_1 f\left(t - \frac{r}{v}\right) + c_2 g\left(t + \frac{r}{v}\right)$$

$$\Phi(r, t) = \frac{c_1 f\left(t - \frac{r}{v}\right)}{r} \xrightarrow{\text{与} t \text{ 无关时}} \Phi(\vec{r}) = \frac{q}{4\pi\epsilon r} \quad \Phi(\vec{r}, t) = \frac{q\left(t - \frac{r}{v}\right)}{4\pi\epsilon r}$$

$$\Phi(\vec{r}, t) = \frac{q \left(t - \frac{r}{v} \right)}{4\pi\epsilon r}$$



$$\Phi(\vec{r}, t) = \frac{q \left(t - \frac{R}{v} \right)}{4\pi\epsilon R}$$



$\Phi(\vec{r}, t)$ 滞后位
场比源滞后时间

$$\Delta t = \frac{R}{v}$$

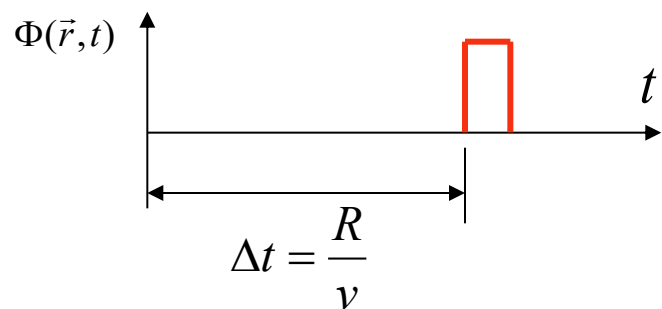
$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

速度

对于连续分布的时变电荷 $\rho(\vec{r}, t)$

$$\Phi(\vec{r}, t) = \iiint_V \frac{\rho \left(\vec{r}', t - \frac{R}{v} \right) dV'}{4\pi\epsilon R}$$

$$\vec{A}(\vec{r}, t) = \frac{\mu}{4\pi} \iiint_V \frac{\vec{J} \left(\vec{r}', t - \frac{R}{v} \right) dV'}{R}$$



当源已消失后,但场还存在

辐射场

5.5 时变电磁场的唯一性定理

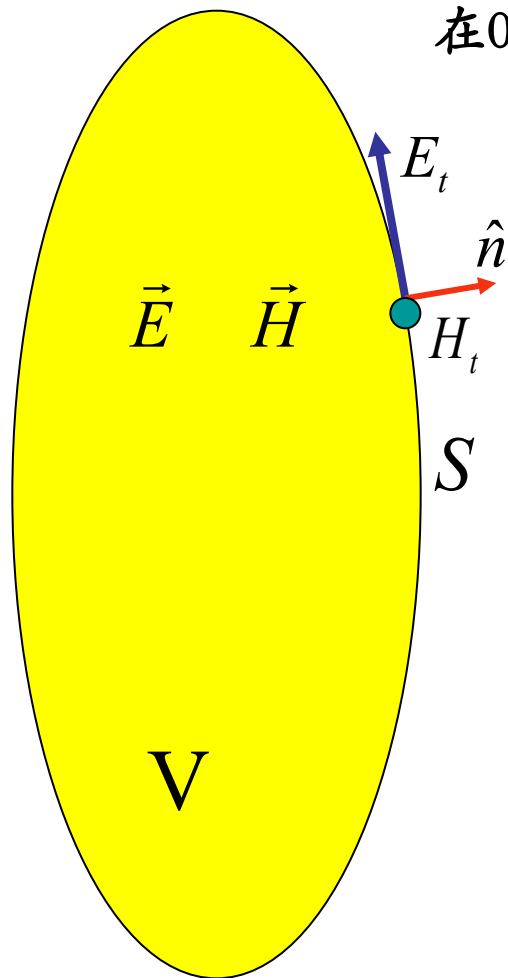
在**闭合面** S 所包围的区域 V 中,
当 **$t=0$ 时刻**的电场强度 E 及磁场强度 H 给定时,
在 $0 < t < T$ 时间内,只要区域 V 边界 S 上**电场强度的切向分量**
或磁场强度的切向分量给定
那么,在 $0 < t < T$ 内的任一时刻,区域中任一点的电磁场
由**Maxwell方程**唯一确定。

时变电磁场的唯一性定理的条件:

初始条件: $t=0$ 时刻的电场强度 E 及磁场强度 H

边界条件: 在 $0 < t < T$ 时间内,区域 V 边界 S 上
电场强度的切向分量
或磁场强度的切向分量

源与场的关系: Maxwell方程



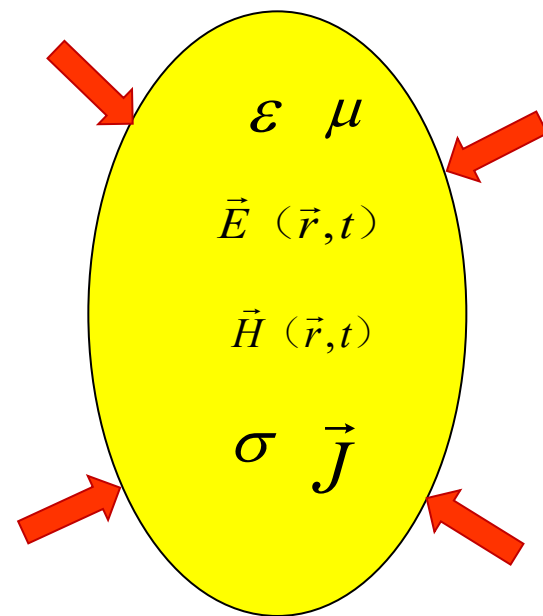
5.6 时变电磁场的能量与功率

1) 时变电磁场的能量

$$\vec{E}(\vec{r}, t) \rightarrow w_e(\vec{r}, t) = \frac{1}{2} \varepsilon |\vec{E}(\vec{r}, t)|^2$$

$$\vec{H}(\vec{r}, t) \rightarrow w_m(\vec{r}, t) = \frac{1}{2} \mu |\vec{H}(\vec{r}, t)|^2$$

$$w(\vec{r}, t) = w_e + w_m$$



2) 损耗功率

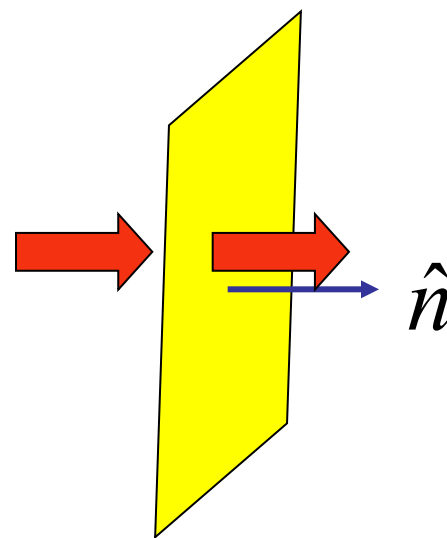
$$\text{损耗功率密度} \quad p(\vec{r}, t) = \sigma |\vec{E}(\vec{r}, t)|^2$$

3) 功率流密度矢量 $\vec{S}$

单位时间垂直穿过单位面积的能量

垂直穿过单位面积的功率

$$P_{in} = - \oint \vec{S} \cdot d\vec{s} = \frac{\partial}{\partial t} (w_e + w_m) + P$$



分析矢量 $\vec{E} \times \vec{H}$ 垂直穿过空间任意封闭面S的通量:

$$\oiint_S \vec{E} \times \vec{H} \cdot d\vec{S} = \iiint_V \nabla \cdot (\vec{E} \times \vec{H}) dV$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot \nabla \times \vec{E} - \vec{E} \cdot \nabla \times \vec{H}$$

$$= \iiint_V \vec{H} \cdot \nabla \times \vec{E} dV - \iiint_V \vec{E} \cdot \nabla \times \vec{H} dV$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

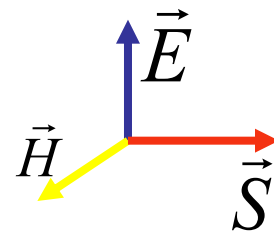
$$= -\iiint_V \vec{H} \cdot \frac{\partial}{\partial t} \mu \vec{H} dV - \iiint_V \vec{E} \cdot \sigma \vec{E} dV - \iiint_V \vec{E} \cdot \frac{\partial}{\partial t} \epsilon \vec{E} dV$$

$$\vec{H} \frac{\partial}{\partial t} \vec{H} = \frac{1}{2} \frac{\partial}{\partial t} \vec{H}^2$$

$$= -\frac{\partial}{\partial t} \left(\iiint_V \frac{1}{2} \mu H^2 dV + \iiint_V \frac{1}{2} \epsilon E^2 dV \right) - \iiint_V \sigma E^2 dV$$

$$\vec{E} \frac{\partial}{\partial t} \vec{E} = \frac{1}{2} \frac{\partial}{\partial t} \vec{E}^2$$

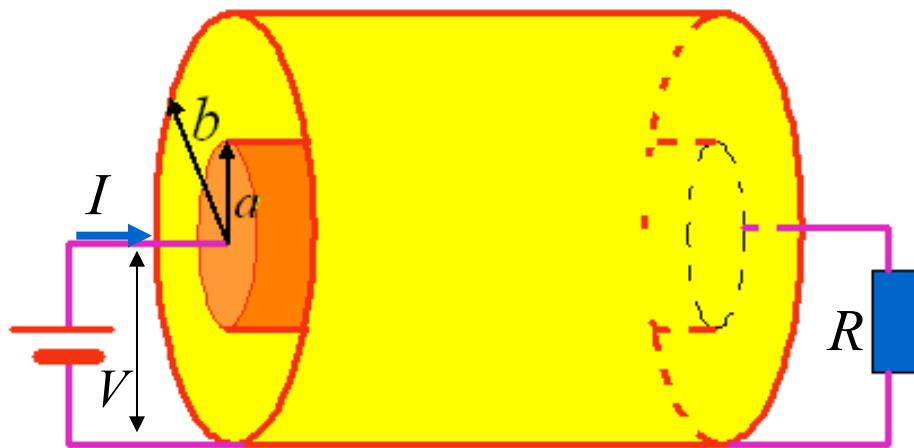
$\vec{S} = \vec{E} \times \vec{H}$ 功率流密度矢量——坡印廷矢量。



例：一同轴线连接电源和负载的电路，

1) 同轴线导体为理想导体

求电源提供给负载的功率。



解：同轴线导体为理想导体

由同轴线内外导体间电压 V

由同轴线内外导体的电流 I

$$\vec{S} = \vec{E} \times \vec{H}$$

在内导体中
 $\rho < a$

$$\vec{E} = 0$$

$$\vec{H} = \hat{\phi} \frac{I\rho}{2\pi a^2}$$

$$\vec{S} = 0$$

在内外导体之间
 $a \leq \rho \leq b$

$$\vec{E} = \hat{\rho} \frac{V}{\ln \frac{b}{a}} \frac{1}{\rho}$$

$$\vec{H} = \hat{\phi} \frac{I}{2\pi\rho}$$

$$\vec{S} = \vec{E} \times \vec{H} = \hat{z} \frac{VI}{2\pi \ln \frac{b}{a}} \frac{1}{\rho^2}$$

在同轴线外
 $\rho > b$

$$\vec{E} = 0$$

$$\vec{H} = 0$$

$$\vec{S} = 0$$

从电源流向负载的功率

$$P = \iint_A \vec{E} \times \vec{H} \cdot d\vec{S} = \int_a^b \frac{VI}{2\pi \ln \frac{b}{a}} \frac{1}{\rho^2} \hat{z} \cdot \hat{z} 2\pi\rho d\rho = VI$$

5.7 正弦电磁场

随时间正弦(余弦)变化的时变电磁场称为正弦电磁场
也称为时谐电磁场

$$\vec{E}(\vec{r}, t) = \vec{E}_m(\vec{r}) \cos(\omega t + \varphi_e)$$

$$\vec{H}(\vec{r}, t) = \vec{H}_m(\vec{r}) \cos(\omega t + \varphi_m)$$

正弦电磁场场强振幅的方向与时间无关，但其大小随时间的变化规律为正弦函数。

正弦电磁场的场和源具有相同的频率。

一、正弦场量的复数形式

- 角频率为 ω 的正弦波，其电场强度可表示为：

$$\overline{E}(\vec{r}, t) = \hat{x}E_{xm}(\vec{r})\cos(\omega t + \varphi_x) + \hat{y}E_{ym}(\vec{r})\cos(\omega t + \varphi_y) + \hat{z}E_{zm}(\vec{r})\cos(\omega t + \varphi_z)$$

$$E_x(\vec{r}, t) = \sqrt{2}E_x(\vec{r})\cos(\omega t + \varphi_x) = \text{Re}[\sqrt{2}E_x(\vec{r})e^{j(\omega t + \varphi_x)}]$$

$$= \text{Re}[\sqrt{2}\boxed{E_x(\vec{r})e^{j\varphi_x}}e^{j\omega t}] = \text{Re}[\sqrt{2}\dot{E}_x(\vec{r})e^{j\omega t}]$$

$$\overline{E}(\vec{r}, t) = \text{Re}[\sqrt{2}e^{j\omega t}(\hat{x}\dot{E}_x(\vec{r}) + \hat{y}\dot{E}_y(\vec{r}) + \hat{z}\dot{E}_z(\vec{r}))]$$
$$\dot{\vec{E}}(\vec{r})$$

$$\overline{E}(\vec{r}, t) = \text{Re}[\sqrt{2}e^{j\omega t}\dot{\vec{E}}(\vec{r})]$$

$$\dot{f}(\vec{r}) = f_0(\vec{r})e^{j\varphi(\vec{r})}$$

$$f(\vec{r}, t) = \text{Re}(\sqrt{2}\dot{f}e^{j\omega t})$$

二、复数场量的性质

$$\overline{A}(\bar{r}, t) = \overline{B}(\bar{r}, t) \quad \Rightarrow \quad \dot{\overline{A}}(\bar{r}) = \dot{\overline{B}}(\bar{r})$$

$$\overline{A}(\bar{r}, t) = \overline{B}(\bar{r}, t) + \overline{C}(\bar{r}, t) \quad \Rightarrow \quad \dot{\overline{A}}(\bar{r}) = \dot{\overline{B}}(\bar{r}) + \dot{\overline{C}}(\bar{r})$$

$$\overline{A}(\bar{r}, t) = f(\bar{r})\overline{B}(\bar{r}, t) \quad \Rightarrow \quad \dot{\overline{A}}(\bar{r}) = f(\bar{r})\dot{\overline{B}}(\bar{r})$$

$$\overline{A}(\bar{r}, t) = \frac{\partial}{\partial x} \overline{B}(\bar{r}, t) \quad \Rightarrow \quad \dot{\overline{A}}(\bar{r}) = \frac{\partial}{\partial x} \dot{\overline{B}}(\bar{r})$$

$$\overline{A}(\bar{r}, t) = \frac{\partial}{\partial t} \overline{B}(\bar{r}, t) \quad \Rightarrow \quad \dot{\overline{A}}(\bar{r}) = j\omega \dot{\overline{B}}(\bar{r})$$

$$\overline{A}(\bar{r}, t) = \int \overline{B}(\bar{r}, t) dt \quad \Rightarrow \quad \dot{\overline{A}}(\bar{r}) = \frac{1}{j\omega} \dot{\overline{B}}(\bar{r})$$

时域中对时间的**求导和积分**
转换为频域的代数运算

$$\frac{d}{dt} \leftrightarrow j\omega$$

$$\int dt \leftrightarrow \frac{1}{j\omega}$$

三、电磁场方程的复数表示

1、麦克斯韦方程

$$\nabla \cdot \bar{D}(\bar{r}, t) = \rho(\bar{r}, t)$$

$$\nabla \cdot \dot{\bar{D}}(\bar{r}) = \dot{\rho}(\bar{r})$$

$$\nabla \times \bar{E}(\bar{r}, t) = -\frac{\partial \bar{B}(\bar{r}, t)}{\partial t}$$

$$\nabla \times \dot{\bar{E}}(\bar{r}) = -j\omega \dot{\bar{B}}(\bar{r})$$

$$\nabla \cdot \bar{B}(\bar{r}, t) = 0$$

$$\nabla \cdot \dot{\bar{B}}(\bar{r}) = 0$$

$$\nabla \times \bar{H}(\bar{r}, t) = \bar{J}(\bar{r}, t) + \frac{\partial \bar{D}(\bar{r}, t)}{\partial t}$$

$$\nabla \times \dot{\bar{H}}(\bar{r}) = \dot{\bar{J}}(\bar{r}) + j\omega \dot{\bar{D}}(\bar{r})$$

$$\oiint_S \bar{D} \cdot d\bar{S} = q$$

$$\oiint_S \dot{\bar{D}} \cdot d\bar{S} = \dot{q}$$

$$\oint_l \bar{E} \cdot d\bar{l} = - \oiint_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{S}$$

$$\oint_l \dot{\bar{E}} \cdot d\bar{l} = -j\omega \oiint_S \dot{\bar{B}} \cdot d\bar{S}$$

$$\oint_l \bar{H} \cdot d\bar{l} = I + \oiint_S \frac{\partial \bar{D}}{\partial t} \cdot d\bar{S}$$

$$\oint_l \dot{\bar{H}} \cdot d\bar{l} = I + j\omega \oiint_S \dot{\bar{D}} \cdot d\bar{S}$$

$$\oiint_S \bar{B} \cdot d\bar{S} = 0$$

$$\oiint_S \dot{\bar{B}} \cdot d\bar{S} = 0$$

$$\bar{D} = \varepsilon \bar{E}$$

$$\bar{B} = \mu \bar{H}$$

$$\bar{J} = \sigma \bar{E}$$

$$\dot{\bar{D}} = \varepsilon \dot{\bar{E}}$$

$$\dot{\bar{B}} = \mu \dot{\bar{H}}$$

$$\dot{\bar{J}} = \sigma \dot{\bar{E}}$$

2、电荷守恒定理

$$\oiint_S \bar{J} \cdot d\bar{S} = -\frac{\partial q}{\partial t}$$

$$\oiint_S \dot{\bar{J}} \cdot d\bar{S} = -j\omega\dot{q}$$

$$\nabla \cdot \bar{J} = -\frac{\partial \rho}{\partial t}$$

$$\nabla \cdot \dot{\bar{J}} = -j\omega\dot{\rho}$$

3、波动方程

$$\nabla^2 \bar{E} - \mu\epsilon \frac{\partial^2}{\partial t^2} \bar{E} = \frac{1}{\epsilon} \nabla \rho + \mu \frac{\partial}{\partial t} \bar{J}$$

$$\nabla^2 \dot{\bar{E}} + \omega^2 \mu\epsilon \dot{\bar{E}} = \frac{1}{\epsilon} \nabla \dot{\rho} + j\omega\mu \dot{\bar{J}}$$

$$\nabla^2 \bar{H} - \mu\epsilon \frac{\partial^2}{\partial t^2} \bar{H} = -\nabla \times \bar{J}$$

$$\nabla^2 \dot{\bar{H}} + \omega^2 \mu\epsilon \dot{\bar{H}} = -\nabla \times \dot{\bar{J}}$$

4、位函数方程

$$\overline{B} = \nabla \times \overline{A}$$

$$\dot{\overline{B}} = \nabla \times \dot{\overline{A}}$$

$$\overline{E} = -\nabla \phi - \frac{\partial}{\partial t} \overline{A}$$

$$\dot{\overline{E}} = -\nabla \dot{\phi} - j\omega \dot{\overline{A}}$$

$$\nabla \cdot \overline{A} = -\mu\epsilon \frac{\partial \phi}{\partial t}$$

$$\nabla \cdot \dot{\overline{A}} = -j\omega\mu\epsilon \dot{\phi}$$

$$\nabla^2 \overline{A} - \mu\epsilon \frac{\partial^2}{\partial t^2} \overline{A} = -\mu \overline{J}$$

$$\nabla^2 \dot{\overline{A}} + \omega^2 \mu\epsilon \dot{\overline{A}} = -\mu \dot{\overline{J}}$$

$$\nabla^2 \phi - \mu\epsilon \frac{\partial^2}{\partial t^2} \phi = -\frac{\rho}{\epsilon}$$

$$\nabla^2 \dot{\phi} + \omega^2 \mu\epsilon \dot{\phi} = -\frac{\dot{\rho}}{\epsilon}$$

$$\bar{A}(\bar{r}, t) = \frac{\mu}{4\pi} \iiint_{V'} \frac{\bar{J}(\bar{r}', t - \frac{R}{v})}{R} dV'$$

$$v = 1 / \sqrt{\mu\epsilon}$$

$$\dot{\bar{A}}(\bar{r}) = \frac{\mu}{4\pi} \iiint_{V'} \frac{\dot{\bar{J}}(\bar{r}') e^{-j\omega \frac{R}{v}}}{R} dV'$$

$$k = \frac{\omega}{v}$$

$$= \frac{\mu}{4\pi} \iiint_{V'} \frac{\dot{\bar{J}}(\bar{r}') e^{-jkR}}{R} dV'$$

$$\phi(\bar{r}, t) = \frac{1}{4\pi\epsilon} \iiint_{V'} \frac{\rho(\bar{r}', t - \frac{R}{v})}{R} dV'$$

$$\dot{\phi}(\bar{r}) = \frac{1}{4\pi\epsilon} \iiint_{V'} \frac{\dot{\rho}(\bar{r}') e^{-jkR}}{R} dV'$$

四、媒质的极化滞后特性

$$\bar{E}(\bar{r}, t) = \sqrt{2} \bar{E}(\bar{r}) \cos(\omega t + \varphi_e)$$

$$\begin{aligned} \bar{P}(\bar{r}, t) &= \varepsilon_0 \chi_e \sqrt{2} \bar{E}(\bar{r}) \cos(\omega t + \varphi_e - \alpha) = \varepsilon_0 \chi_e \operatorname{Re}[\sqrt{2} \bar{E}(\bar{r}) e^{j(\omega t + \varphi_e - \alpha)}] \\ &= \varepsilon_0 \chi_e \operatorname{Re}[\sqrt{2} \dot{\bar{E}}(\bar{r}) e^{j\omega t} e^{-j\alpha}] \end{aligned}$$

$$\dot{\bar{P}}(\bar{r}) = \varepsilon_0 \chi_e \dot{\bar{E}}(\bar{r}) e^{-j\alpha} = \varepsilon_0 \dot{\chi}_e \dot{\bar{E}}(\bar{r}) \quad \dot{\chi}_e = \chi_e e^{-j\alpha}$$

$$\begin{aligned} \varepsilon &= \varepsilon_0 (1 + \dot{\chi}_e) = \varepsilon_0 (1 + \chi_e e^{-j\alpha}) = \varepsilon_0 (1 + \chi_e \cos \alpha - j \chi_e \sin \alpha) \\ &= \varepsilon_0 (\varepsilon'_r - j \varepsilon''_r) = \varepsilon' - j \varepsilon'' \end{aligned}$$

$$\begin{aligned} \nabla \times \dot{\bar{H}} &= \dot{\bar{J}} + j\omega \varepsilon \dot{\bar{E}} = \sigma \dot{\bar{E}} + j\omega \varepsilon \dot{\bar{E}} \\ &= \sigma \dot{\bar{E}} + j\omega (\varepsilon' - j \varepsilon'') \dot{\bar{E}} = (\sigma + \omega \varepsilon'') \dot{\bar{E}} + j\omega \varepsilon' \dot{\bar{E}} \end{aligned}$$

5.8 正弦时变电磁场中的平均能量与功率

1) 平均能量密度

$$w_e(\bar{r}, t) = \frac{1}{2} \varepsilon |\bar{E}(\bar{r}, t)|^2 = \frac{1}{2} \varepsilon \left| \operatorname{Re}[\sqrt{2} \dot{\bar{E}}(\bar{r}) e^{j\omega t}] \right|^2$$

$$= \varepsilon \left| \dot{\bar{E}}(\bar{r}) \right|^2 \cos^2(\omega t)$$

$$\bar{w}_e(\bar{r}) = \frac{1}{T} \int_0^T w_e(\bar{r}, t) dt = \varepsilon \left| \dot{\bar{E}}(\bar{r}) \right|^2 \frac{1}{T} \int_0^T \cos^2(\omega t) dt$$

$$\bar{w}_e(\bar{r}) = \frac{1}{2} \varepsilon \left| \dot{\bar{E}}(\bar{r}) \right|^2 \quad \frac{1}{T} \int_0^T \cos^2 x dx = \frac{1}{2}$$

$$w_m(\bar{r}, t) = \frac{1}{2} \mu |\bar{H}(\bar{r}, t)|^2 \quad \bar{w}_m(\bar{r}) = \frac{1}{2} \mu \left| \dot{\bar{H}}(\bar{r}) \right|^2$$

$$p(\bar{r}, t) = \sigma |\bar{E}(\bar{r}, t)|^2 \quad \bar{p}(\bar{r}) = \sigma \left| \dot{\bar{E}}(\bar{r}) \right|^2$$

2) 平均功率流密度

对于正弦电磁场 设

$$\vec{E}(\vec{r}, t) = \sqrt{2}\vec{E}(\vec{r})\cos(\omega t + \varphi_e) \rightarrow \dot{\vec{E}}(\vec{r}) = \vec{E}(\vec{r})e^{j\varphi_e}$$

$$\vec{H}(\vec{r}, t) = \sqrt{2}\vec{H}(\vec{r})\cos(\omega t + \varphi_h) \rightarrow \dot{\vec{H}}(\vec{r}) = \vec{H}(\vec{r})e^{j\varphi_h}$$

$$\begin{aligned}\vec{S}(\vec{r}, t) &= \vec{E}(\vec{r}, t) \times \vec{H}(\vec{r}, t) = 2\vec{E}(\vec{r}) \times \vec{H}(\vec{r}) \cos(\omega t + \varphi_e) \cos(\omega t + \varphi_h) \\ &= \vec{E}(\vec{r}) \times \vec{H}(\vec{r}) [\cos(2\omega t + \varphi_e + \varphi_h) + \cos(\varphi_e - \varphi_h)]\end{aligned}$$

平均功率流密度

$$\bar{\vec{S}}(\vec{r}) = \frac{1}{T} \int_0^T \vec{S}(\vec{r}, t) dt = \vec{E}(\vec{r}) \times \vec{H}(\vec{r}) \cos(\varphi_e - \varphi_h)$$

$$= \text{Re}[\dot{\vec{E}} \times \dot{\vec{H}}^*]$$

$$\dot{\vec{E}} \times \dot{\vec{H}}^* = \vec{E}(\vec{r}) \times \vec{H}(\vec{r}) e^{j(\varphi_e - \varphi_h)}$$

$$\vec{S}_c(\vec{r}) = \dot{\vec{E}} \times \dot{\vec{H}}^*$$

复Poynting矢量

$$\bar{\vec{S}} = \text{Re}[\vec{S}_c]$$

3) 复Poynting定理 $\vec{S}_c(\vec{r}) = \vec{E} \times \vec{H}^*$

$$\nabla \cdot \vec{S}_c = \nabla \cdot (\vec{E} \times \vec{H}^*) = \vec{H}^* \cdot \nabla \times \vec{E} - \vec{E} \cdot \nabla \times \vec{H}^* \quad (\nabla \times \vec{H})^*$$

$$= \vec{H}^* \cdot (-j\omega\mu\vec{H}) - \vec{E} \cdot (\sigma\vec{E}^* - j\omega\epsilon^*\vec{E}^*)$$

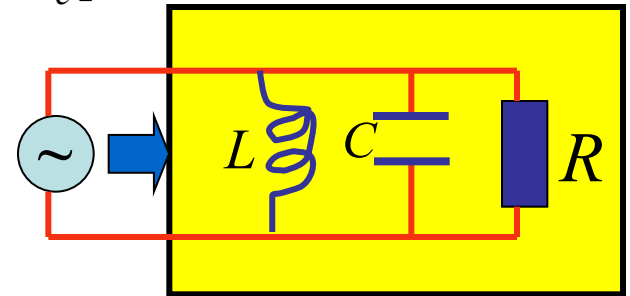
$$= -j2\omega \left[\frac{1}{2} \mu |\vec{H}|^2 - \frac{1}{2} \epsilon^* |\vec{E}|^2 \right] - \sigma |\vec{E}|^2 \quad \epsilon^* = \epsilon' + j\epsilon''$$

$$= -j2\omega \left[\underbrace{\frac{1}{2} \mu |\vec{H}|^2}_{\bar{W}_m} - \underbrace{\frac{1}{2} \epsilon' |\vec{E}|^2}_{\bar{W}_e} \right] - \omega\epsilon'' |\vec{E}|^2 - \sigma |\vec{E}|^2$$

$$\begin{aligned} -\oint_S \vec{S}_c \cdot d\vec{S} &= \iiint_V [\sigma + \omega\epsilon''] |\vec{E}|^2 dV + j2\omega \iiint_V [\bar{W}_m - \bar{W}_e] dV \\ &= \bar{P} + j2\omega(\bar{W}_m - \bar{W}_e) \end{aligned}$$

有功功率 $\bar{P} = \bar{P}_c + \bar{P}_e$

无功功率 $Q = 2\omega(\bar{W}_m - \bar{W}_e)$



$$P + jQ$$

有功功率
无功功率

例： 设 $\vec{E}(\vec{r}, t) = \hat{x}\sqrt{2}E_0 \cos(\omega t)$

$$\vec{H}(\vec{r}, t) = \hat{y}\sqrt{2}H_0 \cos(\omega t + \varphi_0)$$

求： $\vec{S}$ $\vec{S}_c$ $\overline{S}$

解： $\vec{S}(\vec{r}, t) = \vec{E}(\vec{r}, t) \times \vec{H}(\vec{r}, t) = \hat{z}2E_0H_0 \{ \cos(2\omega t + \varphi_0) + \cos \varphi_0 \}$

$$\vec{\dot{E}}(\vec{r}) = \hat{x}E_0 \quad \vec{\dot{H}}(\vec{r}) = \hat{y}H_0 e^{j\varphi_0}$$

$$\vec{S}_c = \vec{\dot{E}} \times \vec{\dot{H}}^* = \hat{z}E_0H_0 e^{-j\varphi_0} = \hat{z}E_0H_0 \{ \cos \varphi_0 - j \sin \varphi_0 \}$$

$$\overline{S}(\vec{r}) = \text{Re}(\vec{S}_c) = \hat{z}E_0H_0 \cos \varphi_0$$

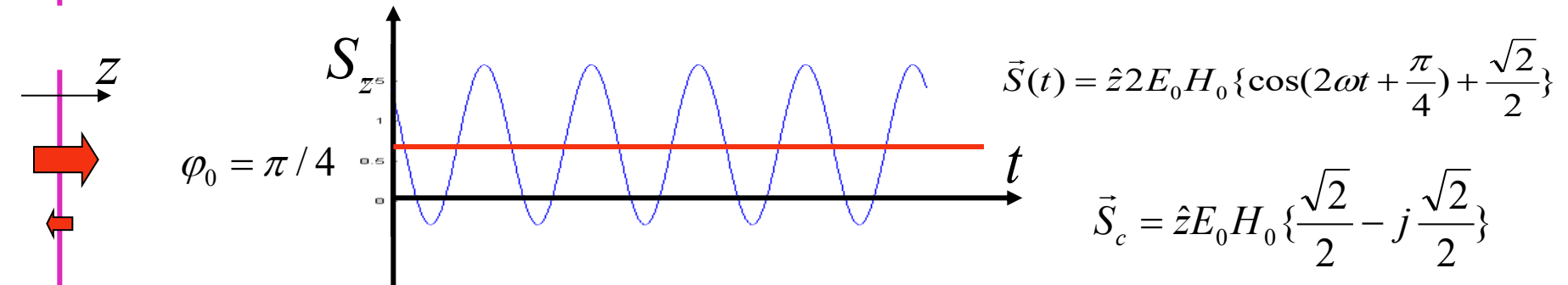
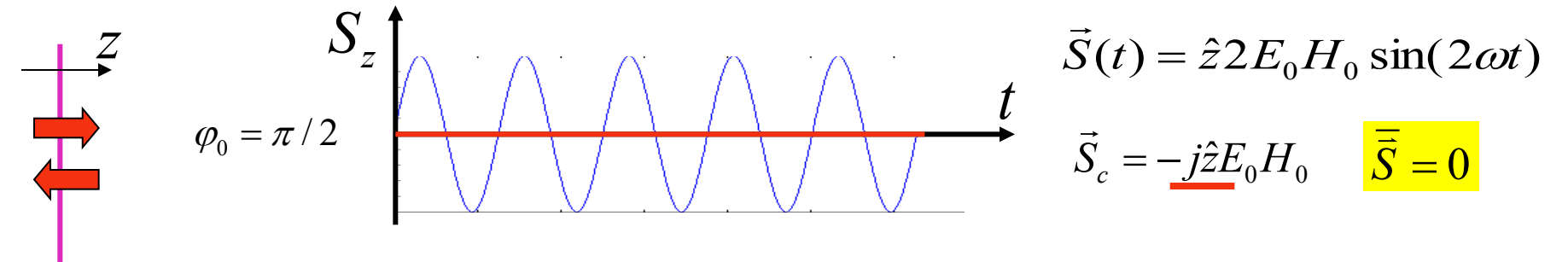
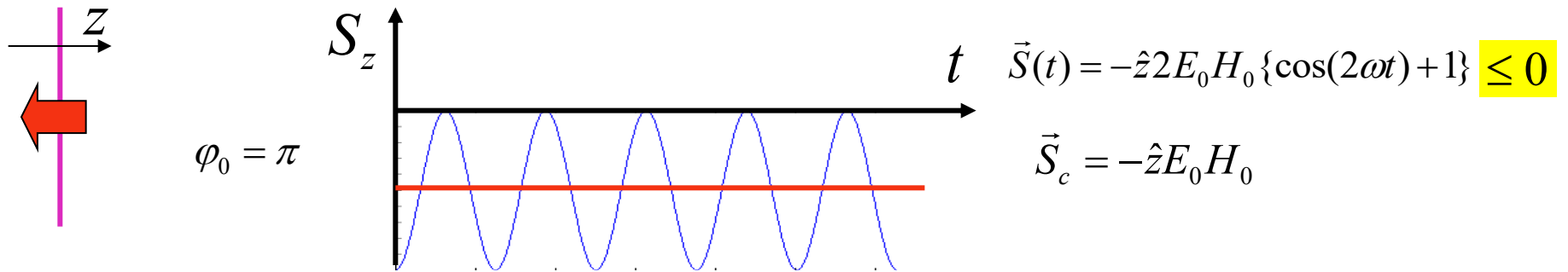
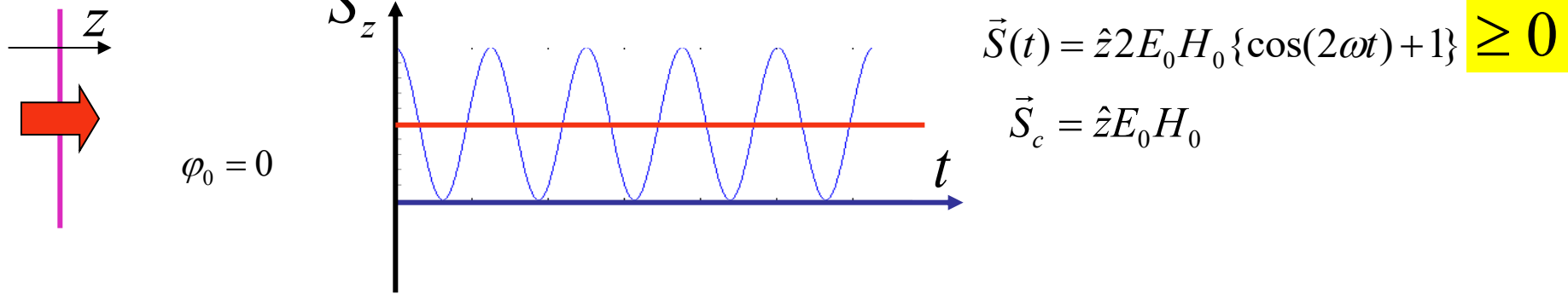
$$\varphi_0 \quad \vec{S}(\vec{r}, t) = \hat{z}2E_0H_0 \{ \cos(2\omega t + \varphi_0) + \cos \varphi_0 \} \quad \vec{S}_c = \hat{z}E_0H_0 \{ \cos \varphi_0 - j \sin \varphi_0 \}$$

$$\varphi_0 = 0 \quad \begin{aligned} &= \hat{z}2E_0H_0 \{ \cos(2\omega t) + 1 \} \\ &= \hat{z}E_0H_0 \end{aligned}$$

$$\varphi_0 = \pi \quad \begin{aligned} &= -\hat{z}2E_0H_0 \{ \cos(2\omega t) + 1 \} \\ &= -\hat{z}E_0H_0 \end{aligned}$$

$$\varphi_0 = \pi/2 \quad \begin{aligned} &= \hat{z}2E_0H_0 \sin(2\omega t) \\ &= -j\hat{z}E_0H_0 \end{aligned}$$

$$\varphi_0 = \pi/4 \quad \begin{aligned} &= \hat{z}2E_0H_0 \left\{ \cos(2\omega t + \frac{\pi}{4}) + \frac{\sqrt{2}}{2} \right\} \\ &= \hat{z}E_0H_0 \left\{ \frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2} \right\} \end{aligned}$$



本章总结

位移电流

时变电磁场的方程(Maxwell方程)

时变电磁场的边界条件

波动方程

矢量磁位和标量电位函数

求位函数方程得到滞后位

时变场的能量与功率,*Poynting*矢量和*Poynting*定理

正弦时变电磁场量和Maxwell方程的复数形式

正弦电磁场的平均能量与功率,复*Poynting*矢量和*Poynting*定理